

ZikaBooking

Product Requirements Document

Worldwide Hotel · Apartment · Car Rental Platform

Version 1.0 — April 2026

Hotels

Apartments

Car Rentals

Worldwide

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1. Platform overview

ZikaBooking is a global travel accommodation and car rental platform that serves customers and suppliers in every country. Users can search, compare, and instantly book hotels, apartments, and cars anywhere in the world, with localised payment methods, multi-currency pricing, and regional support. Born with an African identity, ZikaBooking extends its reach worldwide.

Three booking categories:

- **Hotels** — Named rooms with official star ratings (1–5★), accreditation documents, and manual admin approval before going live.
- **Apartments** — Self-contained units listed by bedroom/bathroom count. Auto-activated on submission. No admin approval gate.
- **Cars** — Individual vehicles or fleet units. Auto-activated. Reactive moderation only.

Key platform capabilities:

- Worldwide search with geo-radius, date-range, and category filters
- Two-way iCal channel manager syncing with Airbnb and Booking.com
- Two payment rails: Tara (mobile money, Africa) and Stripe (global)
- ZIKA-XXXX-CC booking references with atomic DB sequence
- 5-minute reservation lock preventing double-bookings
- Voucher engine with activity scoping, tier locks, and Best Offer banner
- Permanent activity promotion banner with red badge on all listing cards
- Loyalty programme: Bronze → Silver → Gold → Diamond tiers
- Role-based admin panel: 6 roles with country-scoped permissions
- Auto-suspension after 2 consecutive negative reviews (apts & cars)
- Confirmation email + PDF voucher generated and attached at booking
- Multi-currency display with live FX rates; charges in listing base currency

2. User registration & authentication

2.1 Registration flow

When a new user creates an account, the account is created in a **pending_verification** state. No bookings can be made or received until email is verified.

Required fields:

- First name, last name (required)
- Email address (unique, validated format)
- Password (min 8 chars, 1 uppercase, 1 number, 1 special character)
- Account type: Guest or Provider/Host
- Provider only: business name, country

Server-side validation: Email format check, uniqueness check, password strength check. Errors returned field-level, not as a block message. If email already exists: "An account with this email already exists."

2.2 Email verification

A verification email is sent within 5 seconds of account creation via SendGrid.

Token security model:

- Generated: `crypto.randomBytes(32).toString("hex")` — 64-char hex, 256-bit entropy
- Storage: Only SHA-256 hash stored in DB. Plain token sent in email only. Safe if DB is compromised.
- Expiry: 24-hour TTL. Background job purges expired tokens hourly.
- Single-use: `token.used = true` immediately on first successful verification.
- Rate limiting: 10 requests/min per IP on `/verify` · 3 resends/hour per user

Verification link: `https://zikabooking.com/verify?token=<64-char-token>`

On success: `users.status` → `active` · `email_verified` = `true` · user auto signed-in · toast "Email verified — welcome to ZikaBooking!"

On expired link: Guest shown expiry page with "Send a new verification link" CTA.

Resend: Old token invalidated (superseded), new token generated, cooldown 60s, max 3 resends/hour.

2.3 OAuth (Google / Apple)

OAuth users have their email pre-verified by the OAuth provider. Account is created with `status = active` directly. A welcome email (no verification link) is sent instead.

2.4 Account status lifecycle

Status	Description
<code>pending_verification</code>	Created, awaiting email click
<code>active</code>	Email verified — can book and receive bookings
<code>suspended</code>	Policy violation — admin action required
<code>banned</code>	Permanent removal from platform

2.5 Two-factor authentication (admin panel)

All admin panel users require 2FA on every login. TOTP authenticator app for all roles. FIDO2/WebAuthn hardware key required for Super Admin. SMS 2FA not offered. Session tokens expire after 8 hours of inactivity.

3. Listing management

3.1 Hotel requirements

Hotels require manual admin approval before going live. Apartments and cars are auto-activated.

Required fields:

- Name * — full property name
- Room type * — Standard, Superior, Deluxe, Suite, Junior Suite, Studio, Family Room, Presidential Suite
- Number of units * — integer ≥ 1 (rooms of this type available)
- Price per night * — decimal
- Currency — defaults to user locale
- Address * — Google Places Autocomplete + geocoding → map pin auto-update
- Town / city * — auto-filled from geocoding, editable
- Country — auto-filled from geocoding
- Services / amenities — multi-select checkbox grid (see section 3.4)
- Accreditation documents — business licence, hotel operating permit, tourism authority certificate (PDF/image, required for approval)
- Star rating — assigned by admin only (1★–5★); provider may indicate claimed rating
- Photos — up to 30 images, 5 MB each, JPEG/PNG/WEBP; first = cover
- Description — free text, max 1 000 chars
- Check-in / check-out times, cancellation policy (Flexible/Moderate/Strict)
- Minimum stay (nights), smoking allowed, pets allowed

Validation workflow: draft → pending_review → approved / rejected. Only approved hotels appear in search. Rejected hotels can resubmit; each resubmission creates a new review task.

3.2 Apartment requirements

Required fields:

- Name * — listing title
- Address * — Google Places Autocomplete + geocoding → live map pin
- Town / city * — auto-filled from geocoding, editable
- Country — auto-filled from geocoding
- Bedrooms, bathrooms, max guests — integer counters
- Price per night *, currency
- Services / amenities — same multi-select grid as hotels
- Photos — up to 30 images; first = cover
- Description — max 1 000 chars
- Check-in / check-out times, cancellation policy, minimum stay
- Smoking allowed, pets allowed, long-stay discount toggle

Activation: Auto-activated on submission once required fields and ≥ 3 photos pass basic validation. No admin review gate.

3.3 Car rental requirements

Identity & classification:

- Listing title *, Make * (brand), Model *, Year * (1990–current)
- Category * — Economy/Compact/SUV/Minivan/Pickup/Luxury/Electric/Convertible
- Number of units * — fleet count
- Colour (palette), Licence plate * (private), Odometer reading *

Technical specs:

- Seats *, Doors *, Transmission * (Manual/Automatic/Semi-auto)
- Fuel type * (Petrol/Diesel/Hybrid/Electric/LPG), Drive type (2WD/4WD/AWD)
- Engine size (optional), Air conditioning toggle *

Rental terms:

- Price per day *, Currency, Minimum rental days, Minimum driver age (default 21)
- Mileage policy * (Unlimited / Limited — if limited: km/day + extra km rate)
- Fuel policy * (Full-to-Full / Full-to-Empty / Pre-purchase)
- Security deposit, Cancellation policy

Insurance & documents:

- Insurance type * (Basic 3rd party / Comprehensive / Premium zero-excess)
- Roadside assistance toggle, Cross-border allowed toggle
- Vehicle registration upload * (PDF/image), Insurance certificate *, Roadworthiness cert (optional)

Pickup & return:

- Pickup address * + geocoding → map pin, Town *, Country
- Return same location toggle, Airport pickup toggle
- Delivery toggle (+ radius km + fee if enabled), Pickup hours (from / to)

Activation: Auto-activated on submission. Reactive moderation only.

3.4 Services & amenities grid

Both hotels and apartments share the same multi-select amenity grid, grouped by category:

Category	Amenities
Connectivity	WiFi, Smart TV, Work desk, Printer
Food & drink	Breakfast included, Restaurant on-site, Coffee machine, Minibar, Kitchen / kitchenette
Wellness	Swimming pool, Fitness centre, Spa, Sauna, Hot tub
Comfort	Air conditioning, Heating, Laundry, Parking, Elevator, Accessible
Services	24h reception, Daily housekeeping, Airport shuttle, 24h security, Shop on-site, Pet-friendly

Providers can also add custom services via a free-text field. Custom items appear as chips in the guest-facing listing.

3.5 Geocoding & live map pin

When the provider types or selects an address, the platform calls the Google Geocoding API, extracts lat, lng, locality (→ Town), and country, auto-fills those fields, and drops a draggable pin on the embedded map. If the provider drags the pin manually, lat/lng update but the address fields stay. This is the source of truth for all search radius queries.

3.6 Channel manager — iCal two-way sync

Each ZikaBooking listing generates a unique iCal feed URL. Hosts paste this into Airbnb and Booking.com's import field, and paste those platforms' export URLs into ZikaBooking.

Sync behaviour:

- ZikaBooking polls external feeds every 15 minutes
- Dates in external feeds → blocked in ZikaBooking availability DB
- ZikaBooking confirmed bookings → VEVENT BUSY in emitted .ics feed
- UID field used as idempotency key — re-importing same feed never double-counts
- Backoff on failure: 1 min → 5 min → 15 min → alert after 3 consecutive failures
- Phase 2: Airbnb Connect API + Booking.com Connectivity API (push-based, <1 min latency)

Host dashboard: Channels panel shows all connected OTAs, status (synced/error/pending), last sync timestamp, blocked dates imported. Dates held pending payment shown as amber "Held" — distinct from confirmed bookings (green) and blocked dates (grey).

4. Booking engine

4.1 5-minute reservation lock

When a guest taps "Book Now", the unit is locked for exactly 5 minutes. No other guest can book the same unit for the same dates during this window. If payment is not completed in time, the lock expires, inventory is restored, and the guest is notified.

Lock creation: Triggered on "Book Now" tap only — not at search, not at date selection. Uses database-level row lock or atomic Redis SETNX. Decrements available count by 1 for listings with unit_count > 1.

Timer UX:

- Green (5:00–2:00): "Booking held — complete payment"
- Amber (1:59–0:31): "Hurry — only X:XX remaining!" — urgency copy
- Red + pulse animation (0:30–0:00): "Less than 30 seconds — pay now!"
- Timer synced to server-side expires_at — not paused on tab switch or mobile background

On expiry: Lock status → expired. Inventory restored. Guest sees expiry modal with "Try to rebook" and "Search again" CTAs. Provider and active search results updated within 5 minutes (cache TTL).

On successful payment: Lock status → completed. Booking record created. Inventory permanently decremented.

Guest cancel: If guest clicks "Cancel booking", lock status → released immediately. No wait for natural expiry.

Race condition grace window: If payment initiated before expiry but webhook arrives up to 2 minutes after expiry, booking is honoured. Beyond 2 minutes, payment is refunded.

Idempotency: If same guest refreshes payment page, existing active lock is reused — timer continues from where it started.

4.2 Booking reference format

Format: ZIKA-XXXX-CC

Segment	Description	Example
ZIKA	Brand prefix — fixed, always uppercase	ZIKA
XXXX	Sequential number — zero-padded to 4 digits; grows beyond 4 once > 9999	0001 · 0042 · 10000
CC	ISO 3166-1 alpha-2 country code of the property/pickup location	KE · NG · FR · GB

Generation rules: Reference created at the exact moment payment success webhook is received and validated. DB sequence (NEXTVAL) called inside booking creation transaction — guarantees uniqueness even under high concurrency. Stored as VARCHAR(20) with UNIQUE constraint. Immutable once assigned. Partial search supported (enter "0042" to find across all countries).

5. Payment system

5.1 Tara — mobile money (Africa)

Tara is a unified mobile money aggregator API covering the major African mobile money operators through a single integration.

Supported operators:

Operator	Countries	Currencies
M-Pesa	Kenya, Tanzania, DRC, Mozambique	KES, TZS, MZN, CDF
MTN MoMo	Uganda, Ghana, Rwanda, Cameroon, Côte d'Ivoire	UGX, GHS, RWF, XAF, XOF
Orange Money	Senegal, Mali, Burkina Faso, Madagascar	XOF, MGA
Airtel Money	Zambia, Malawi, Niger	ZMW, MWK
Wave	Senegal, Côte d'Ivoire	XOF
MoMo SA	South Africa	ZAR

Guest flow: Select "Mobile money" → enter number → STK push sent → guest approves on handset → Tara webhook confirms → booking confirmed. Completes in under 60 seconds.

Host payouts: Disbursed via Tara to registered mobile money number. Triggered T+24h after guest check-in.

Technical: STK push timeout 60s. Exponential backoff on failure. Idempotency key = booking reference + attempt number. Webhook endpoint validates HMAC signature.

5.2 Stripe — global payments

Payment methods: Visa, Mastercard, Amex, UnionPay · Apple Pay · Google Pay · PayPal · SEPA/BACS/ACH bank debit · Klarna/Afterpay BNPL · 135+ currencies.

Guest flow: Stripe Elements renders PCI-compliant card input (or wallet button) in an iframe → PaymentIntent created server-side → guest completes → Stripe webhook confirms → booking confirmed. 3D Secure / SCA handled automatically.

Host payouts: Stripe Connect Express accounts. Bank account or debit card. Transfer created T+24h after check-in. Stripe handles currency conversion.

Security: No raw card data on ZikaBooking servers. All Stripe keys in environment variables. publishable key client-side only. secret key server-side only.

5.3 Currency detection & payment selection

On page load, ZikaBooking detects the guest's country via IP geolocation. African countries → Tara recommended first. All others → Stripe methods first. Guest can always switch. Multi-currency display uses live FX rates (updated hourly); actual charge is always in the listing's base currency.

5.4 Billing calculation

Line item	Formula
Base amount	nights (or days) × unit_price
Promotion discount	– base_amount × promotion_rate (if active promotion)

Voucher discount	– calculated amount (if personal voucher applied)
Note on discounts	Best of promotion vs voucher wins. They do NOT stack.
Subtotal	base_amount – discount
Service fee (5%)	CEILING(subtotal × 0.05, 2 decimal places)
Taxes	subtotal × tax_rate (by country: KE/NG/GH 16%, FR/UK 20%, UAE 5%, ZA 10%)
Total	subtotal + service_fee + taxes

Rounding: All monetary values stored as DECIMAL(10,2). Service fee rounded up (CEILING). FX display is informational only.

5.5 Confirmation email & PDF voucher

Triggered: Immediately on payment success webhook. Sent via SendGrid. Auto-retry on failure: 5 min → 30 min → admin alert after 3 failures.

Guest email: Subject: "Your booking is confirmed — ZIKA-XXXX-CC". Contains: booking details, full itemised receipt (base amount, discount line, subtotal, service fee, taxes, total), payment method, transaction ID, cancellation policy reminder, host contact, PDF voucher attached.

Host email: Subject: "New booking received — ZIKA-XXXX-CC". Contains: guest name, dates, net payout amount, payout schedule (T+24h after check-in).

PDF voucher: A4 document. Sections: header (ZikaBooking logo + reference), CONFIRMED badge, guest details, booking details, itemised receipt, payment info, cancellation policy, QR code (encodes booking URL). File name: ZikaBooking-ZIKA-XXXX-CC.pdf. Stored in S3. Attached directly (not a link).

6. Vouchers & promotions

6.1 Voucher types — by activity

Vouchers are scoped to one or more booking categories. A hotel voucher cannot be applied to a car rental booking.

Activity scopes: Hotels only · Apartments only · Cars only · Hotels + Apartments · All categories (universal).

Voucher attributes: code (unique, uppercase, 6–12 chars) · title · description (max 120 chars) · activity_scope · discount_type (percentage or fixed) · discount_value · minimum_booking_value · maximum_discount_cap · valid_from / valid_until · usage_limit_total · usage_limit_per_guest (default 1) · status (active/paused/expired/exhausted) · applicable_tiers · country_scope (null = worldwide).

Application: Guest types code or selects from wallet. Backend validates: code exists → status active → within dates → usage not exhausted → scope matches → tier met → minimum value met. Voucher not consumed until payment confirmed.

6.2 Best Offer banner

When a guest has at least one active, valid, applicable voucher, a Best Offer banner activates across the app.

Banner states:

- Active — full colour, animated shimmer, CTA "Tap to apply"
- Expiring soon (≤ 48 h remaining) — amber pulse, "Offer expires soon!"
- Applied — green confirmation, "Offer applied — saving \$X"
- No voucher — banner hidden entirely (no empty promotional noise)

Personal banner: Dismissable. Tied to guest account. Shows highest-value applicable voucher as headline.

6.3 Tier-based voucher activation

Vouchers can be restricted to specific loyalty tiers. Tier is evaluated at redemption time, not at issuance.

Tier	Points required	Colour
Bronze	0 – 499 pts	#CD7F32
Silver	500 – 1 999 pts	#C0C0C0
Gold	2 000 – 4 999 pts	#D4A017
Diamond	5 000+ pts	#4FC3F7

Tier modes: "Minimum tier" (Gold voucher available to Gold and Diamond). "Exact tier" (Silver-exclusive). "All tiers" (no restriction).

Auto-assignment: When a guest crosses a tier threshold, any active vouchers with applicable_tiers containing that tier and auto_assign = true are added to their wallet instantly with a push notification.

6.4 Activity promotions — permanent banner

Admins can activate a promotional campaign directly on a specific activity category. This triggers a permanent, non-dismissable Best Offer banner for ALL guests browsing that category.

Promotion fields: activity (hotel/apartment/car) · label_text (max 6 chars, e.g. "30%") · label_colour (default red #C84B2F) · discount_type (percentage/fixed/label only) · discount_value · valid_from/valid_until · apply_to_booking toggle · banner_title · banner_subtitle · status (scheduled/active/paused/expired/superseded) · country_scope.

Permanent banner behaviour: Non-dismissable. Shown to all guests in red (#C84B2F). Always shows label prominently (e.g. "30% OFF HOTELS"). Takes visual priority over personal voucher banner.

Red badge on listing cards: Every listing card in the promoted category displays a red badge in the top-left corner showing the label_text. Appears on search results, explore grid, home screen cards, and detail page hero.

One active per category: Only one promotion can be active per activity at a time. Replacing an active promotion sets the old one to "superseded".

Discount precedence: Activity promotion vs personal voucher — higher value wins. No stacking. Applied automatically at checkout when apply_to_booking = true.

7. Reviews & moderation

7.1 Review system

Eligibility: Only guests with a completed, checked-out booking can leave a review. One review per booking. 14-day window after checkout.

Content: Overall star rating (1–5★) + written comment. Host can reply to any review. Both ratings (guest rates listing, host rates guest) are mutual.

Fraud safeguards:

- Verified stays only — non-guests cannot review
- New account detection — both reviews from accounts created within 7 days → flagged, no auto-suspension, fraud alert to admin
- Device fingerprint check — same device or IP subnet → reviews flagged, no auto-suspension

7.2 Auto-suspension on consecutive negative reviews

Applies to APARTMENTS and CARS only. Does NOT apply to hotels (already under manual accreditation).

Rule: 2 straight negative reviews (back-to-back, no positive in between) → listing auto-suspended immediately after the second.

Negative definition: Overall star rating of 1★ or 2★. Ratings of 3★, 4★, 5★ are NOT negative.

Consecutive detection: Counter increments on each 1★ or 2★. Any 3★+ resets counter to zero. Block triggered when counter reaches 2.

On trigger: status → auto_suspended. Listing removed from search. Active reservation locks cancelled, guests notified, payments voided. Provider notified by email. Review task created in admin unblock queue.

7.3 Agent unblock process

Who can unblock: Support Agent, Country Manager (own countries), Admin, Super Admin.

Decision options:

- Unblock — warning issued: listing reactivated, formal warning email sent to provider, counter reset to 0
- Unblock — no warning: reviews deemed unfair/fraudulent, listing reactivated, mandatory internal note required
- Keep suspended — request response: message sent to provider, 72h window, listing stays down
- Permanently ban listing: listing permanently banned, provider notified, Admin/Super Admin only to reverse

SLA: Agent must action within 48 hours. Auto-escalation on breach: support → country manager → admin → super admin.

Escalation on repeat offence: 2nd suspension → escalated to Admin (24h SLA). 3rd suspension → Super Admin only. 3+ listings from same provider suspended → full account review.

8. Admin panel

8.1 Roles & permissions

Six roles exist on the platform. Each user has exactly one role. Roles cannot be stacked.

Feature	Super Admin	Admin	Country Mgr	Sales	Support	Finance
Approve / reject hotel	✓	✓	scope	—	—	—
Assign star rating	✓	✓	scope	—	—	—
Suspend any listing	✓	✓	scope	—	—	—
View all bookings	✓	✓	scope	scope	✓	—
Create manual booking	✓	✓	✓	✓	—	—
Issue refund	✓	✓	scope	—	—	—
View payment records	✓	✓	scope	—	status	✓
Export financial data	✓	—	—	—	—	✓
Manage provider accounts	✓	✓	scope	—	—	—
Create admin users	✓	—	—	—	—	—
Create country managers	✓	—	—	—	—	—
Global platform settings	✓	—	—	—	—	—
View all reports	✓	✓	scope	—	—	finance
Audit log access	✓	✓	—	—	—	—

scope = accessible for own assigned countries only

8.2 Hotel accreditation workflow

Every hotel submission lands in the Hotel review queue. Admins can open uploaded documents in a built-in viewer. Star rating is assigned by the admin only.

- Admin opens listing from queue: sees provider name, submission date, claimed star rating, uploaded documents
- Document viewer: business registration, hotel operating permit, tourism authority certificate, classification evidence
- Star rating assignment: admin selects verified rating (1★–5★). If submitted documents support a different rating than claimed, admin sets the correct one with a note.
- Approve & publish: listing status → active, provider notified, appears in search within 5 minutes
- Reject: admin selects reason (Insufficient documentation / Operating permit expired / Star rating unverifiable / Other) + optional free-text. Provider notified with resubmission instructions.
- Audit trail: every review action logged (admin ID, timestamp, decision, reason, star rating). Immutable.

8.3 Country manager scoping

A country manager's account has a `country_scope` array (e.g. ["KE","TZ","UG"]). Every database query they trigger is filtered at the API level — not just the UI. Even direct API calls outside scope return 403 Forbidden. Super Admin can assign or modify country scope.

8.4 Audit log

Every action in the admin panel is written to an immutable audit log: user_id, role, action, target_type, target_id, old_value, new_value, ip_address, timestamp. Retained 7 years. Searchable by Admin and Super Admin. Export to CSV for Super Admin only.

9. Loyalty programme

ZikaBooking AfriPoints are earned on every confirmed booking and redeemable for discounts on future bookings.

Earning rate: 1 point per \$1 spent (on total amount paid).

Tier	Points required	Bonus multiplier	Key benefit
Bronze	0 – 499 pts	1× (base rate)	Starting tier · all-tier vouchers
Silver	500 – 1 999 pts	1.15× (+15%)	Silver+ vouchers unlocked
Gold	2 000 – 4 999 pts	1.25× (+25%)	Gold+ vouchers · priority support
Diamond	5 000+ pts	1.40× (+40%)	All vouchers · early access · free upgrades

Auto-assignment on tier upgrade: When a guest crosses a tier threshold, any active vouchers with `auto_assign = true` and `applicable_tiers` containing that tier are added to their wallet instantly. Push notification sent: "You've reached Gold! Your exclusive voucher has been added."

Redemption: Points redeemable for fixed discounts at checkout. Minimum redemption threshold configurable per campaign.

10. Channel manager

The channel manager prevents double-bookings across ZikaBooking, Airbnb, and Booking.com using the iCal protocol (RFC 5545) as the universal sync standard.

Host setup — two steps:

- 1. Host copies ZikaBooking export URL (`zikabooking.com/cal/{listing-id}.ics`) and pastes into Airbnb and Booking.com "Import calendar" field.
- 2. Host copies Airbnb and Booking.com export URLs and pastes into ZikaBooking import fields.

Technical requirements:

- Full RFC 5545 compliance: parse and emit VEVENT, DTSTART, DTEND, SUMMARY, UID, STATUS
- Normalise all timestamps to UTC internally
- UID field = idempotency key — re-importing same feed never creates duplicate blocks
- Polling every 15 minutes per connected channel feed
- Exponential backoff on failure: 1 min → 5 min → 15 min → alert after 3 consecutive failures
- Host dashboard: connected OTAs, status (synced/error/pending), last sync timestamp, blocked dates count
- Manual calendar override: hosts can block/unblock date ranges (personal use, maintenance)
- Sync log: 90-day retention of all sync events
- Car rental: same system applies per vehicle

Phase 2 — Direct API: Airbnb Connect API + Booking.com Connectivity API (push-based, <1 min latency). Requires platform certification. Architecture designed so poller swaps to webhook receiver without other changes.

11. Platform targets — Android, iOS & web app

ZikaBooking is a cross-platform product delivered simultaneously as a native Android application, a native iOS application, and a progressive web application (PWA). All three surfaces share a single backend API and a single business logic layer. UI is built with React Native (mobile) and Next.js (web), maximising code reuse while respecting platform conventions.

11.1 Android application

Distribution: Google Play Store (primary) · APK direct-install available for markets where Play Store access is restricted (some African countries).

Minimum SDK: Android 8.0 (API level 26) · Target SDK: latest stable Android release.

Build toolchain: React Native with Expo managed workflow · Gradle build system · ProGuard/R8 for release minification.

Android-specific features:

- Material Design 3 component guidelines followed where React Native defaults diverge
- Back button / gesture navigation handled correctly throughout all screens
- Android App Links (zikabooking.com/booking/ZIKA-XXXX-CC) deep-link into booking detail screen
- Firebase Cloud Messaging (FCM) for push notifications — booking confirmations, timer alerts, voucher auto-assign, payout sent
- Tara STK push: Android receives system-level USSD/STK notification from mobile operator — no in-app overlay needed
- Biometric authentication (fingerprint / face unlock) for returning user sign-in via Android BiometricPrompt API
- Offline mode: last-viewed listing and active booking details cached via AsyncStorage for low-connectivity markets
- Dark mode: follows system Android 10+ dark theme setting automatically
- Google Pay integration via Stripe Android SDK — single-tap payment at checkout
- Google Maps SDK for Android: embedded map with draggable pin on listing creation and detail pages
- Camera access: photo upload for listing creation and ID document upload (car KYC)
- Network security config: certificate pinning on API and payment endpoints
- Play Integrity API: device attestation check before processing payments to prevent fraud on rooted devices

Permissions requested: INTERNET · ACCESS_NETWORK_STATE · CAMERA (listing photos, ID upload) · READ_EXTERNAL_STORAGE / READ_MEDIA_IMAGES (photo picker) · RECEIVE_BOOT_COMPLETED (reschedule local notifications after reboot) · USE_BIOMETRIC · POST_NOTIFICATIONS (Android 13+).

Release pipeline: GitHub Actions CI/CD · Fastlane for build signing and Play Store upload · Internal test track → Closed testing (beta) → Production staged rollout (10% → 50% → 100%).

11.2 iOS application

Distribution: Apple App Store (primary). TestFlight for beta distribution.

Minimum OS: iOS 15.0 · iPadOS 15.0. Target: latest stable iOS release.

Build toolchain: React Native with Expo managed workflow · Xcode build system · Metro bundler · App Store Connect for distribution.

iOS-specific features:

- Apple Human Interface Guidelines followed for navigation patterns, typography, and spacing

- Universal Links (zikabooking.com/booking/ZIKA-XXXX-CC) deep-link into booking detail screen via apple-app-site-association file
- Apple Push Notification service (APNs) via Firebase Cloud Messaging bridge — all push categories identical to Android
- Apple Pay integration via Stripe iOS SDK — Touch ID / Face ID authorisation at checkout
- Sign in with Apple: mandatory OAuth option per App Store guidelines (presented alongside Google Sign-In)
- Face ID / Touch ID for returning user sign-in via LocalAuthentication framework
- SF Symbols used for system-consistent iconography where React Native icon library does not provide platform-native variants
- Dark mode: follows iOS system appearance (light/dark/auto) via React Native Appearance API
- Google Maps SDK for iOS: embedded map with draggable pin (falls back to Apple Maps if Google Maps not available)
- Camera and photo library access via react-native-image-picker: listing photos, ID documents
- iOS App Clips: lightweight entry point for the booking confirmation screen — guest can view booking status and QR code without installing the full app
- Widget extension (iOS 16+ WidgetKit): "My next trip" widget on Home Screen showing upcoming check-in date and listing name
- Handoff: booking research started on iPhone continues on iPad or Mac without re-authentication
- Privacy manifest (PrivacyInfo.xcprivacy): declares all API usage reasons per Apple's App Privacy requirements

Permissions requested: NSCameraUsageDescription (listing photos, ID upload) · NSPhotoLibraryUsageDescription (photo picker) · NSFaceIDUsageDescription (biometric sign-in) · NSLocationWhenInUseUsageDescription (map-based search) · NSUserNotificationsUsageDescription (push notifications).

App Store guidelines compliance: No external payment links (Apple IAP not applicable — ZikaBooking charges are for physical services, exempt under App Store Review Guideline 3.1.1). Age rating: 4+ (no user-generated content concerns). Privacy nutrition label completed for all collected data categories.

Release pipeline: GitHub Actions CI/CD · Fastlane match for certificate management · TestFlight internal → external beta → App Store phased release.

11.3 Web application (PWA)

Framework: Next.js 14+ (App Router) · React 18 · TypeScript · Tailwind CSS.

Rendering strategy:

- Search results pages: Server-Side Rendering (SSR) for SEO — listing names, prices, locations indexed by Google
- Listing detail pages: Static Site Generation (SSG) with Incremental Static Regeneration (ISR, 5-minute revalidation) — fast initial load, fresh availability data
- Booking / payment flow: Client-Side Rendering (CSR) — no sensitive booking state in HTML source
- Admin panel: CSR behind auth middleware — never server-rendered to avoid accidental data exposure

Progressive Web App (PWA) features:

- Web App Manifest (manifest.json): name "ZikaBooking", short_name "Zika", theme_color #C84B2F, background_color #FAFAF7, display: standalone, icons at 192px and 512px
- Service Worker (via next-pwa): precaches shell, homepage, and active booking page for offline access. Network-first strategy for API calls; cache-first for static assets.
- "Add to Home Screen" prompt (beforeinstallprompt) shown after second visit or after completing a booking, whichever comes first

- Offline fallback page shown when user navigates to uncached route without network — shows active booking details from cache
- Web Push Notifications via Web Push API + VAPID keys: booking confirmations, 5-minute timer alerts, voucher expiry warnings. Requires explicit permission grant (browser prompt).
- Background Sync API: if guest submits a booking form while briefly offline, the request is queued and replayed when connectivity is restored

SEO requirements:

- Server-rendered metadata (title, description, Open Graph, Twitter Card) per listing and per city landing page
- Structured data (JSON-LD): LodgingBusiness schema for hotels/apartments, Offer schema for pricing, BreadcrumbList for navigation
- Sitemap.xml auto-generated from active listings (updated nightly)
- robots.txt: allows all listing pages, disallows /admin, /booking/*, /api/*
- Core Web Vitals targets: LCP < 2.5s, FID < 100ms, CLS < 0.1 — monitored via Vercel Analytics / Google Search Console

Browser support: Chrome 90+ · Firefox 90+ · Safari 15+ · Edge 90+. IE11 not supported. Responsive layout: 320px (small Android) → 768px (tablet) → 1280px+ (desktop). Mobile-first CSS breakpoints.

Web-specific payment: Stripe.js loaded from js.stripe.com (never self-hosted). Stripe Elements card input in iframe. Google Pay and Apple Pay rendered via Stripe Payment Request Button API (browser-native wallet detection).

Hosting: Vercel (Next.js optimised) or AWS Amplify. Global CDN edge network. Automatic HTTPS. Preview deployments for every pull request.

11.4 Shared codebase strategy

The three platforms share as much code as possible through a monorepo structure, with platform-specific adaptations isolated in thin platform layers.

Layer	Shared across all platforms	Platform-specific
Business logic	Booking engine, billing calc, voucher validation, review rules, notifications	None — 100% shared
API client	REST/GraphQL queries, response types, error handling, retry logic	None — 100% shared
State management	Zustand stores (listings, booking, auth, vouchers)	None — 100% shared
UI components	~80% shared via React Native Web (works on iOS, Android, web)	~20% platform-specific (navigation, modals, maps, payment buttons)
Navigation	Screen hierarchy and route definitions	React Navigation (mobile) vs Next.js App Router (web)
Payments	Payment intent creation, billing calc, voucher application logic	Carra STK (mobile only) · Apple Pay (iOS + Safari) · Google Pay (Android)
Push notifications	Notification content, deep-link schemas, permission request flow	Firebase (Android) · APNs (iOS) · Web Push / VAPID (web)
Maps	Geocoding API calls, pin logic, address auto-fill	Google Maps SDK (mobile) · Google Maps JS API (web)
Offline	Booking cache schema, cache invalidation rules	AsyncStorage (mobile) · Service Worker + Cache API (web)

11.5 Platform feature matrix

Feature	Android	iOS	Web (PWA)
Full booking flow (search → payment)	Yes	Yes	Yes
Provider listing creation	Yes	Yes	Yes
iCal channel manager dashboard	Yes	Yes	Yes
Admin panel	No — web only	No — web only	Yes

Push notifications	Yes (FCM)	Yes (APNs)	Yes (Web Push, opt-in)
Biometric sign-in	Yes (fingerprint/face)	Yes (Face/Touch ID)	No
Apple Pay	No	Yes (iOS + Safari)	Yes (Safari only)
Google Pay	Yes	No	Yes (Chrome)
Tara mobile money	Yes	Yes	Yes (redirects to USSD)
Offline access (cached booking)	Yes (AsyncStorage)	Yes (AsyncStorage)	Yes (Service Worker)
"Add to Home Screen"	Yes (PWA prompt or Play Store)	Yes (App Store)	Yes (PWA install prompt)
Deep links	Yes (App Links)	Yes (Universal Links)	Yes (URL routing)
iOS App Clips	No	Yes	No
Home screen widget	No (planned Phase 2)	Yes (WidgetKit iOS 16+)	No
SEO-indexed listing pages	No	No	Yes (SSR/SSG)
Structured data (JSON-LD)	No	No	Yes
PDF voucher download	Yes	Yes	Yes
QR code scan at check-in	Yes (camera)	Yes (camera)	Yes (camera API)
Dark mode	Yes (system)	Yes (system)	Yes (prefers-color-scheme)

11.6 App store & deployment requirements

Google Play Store:

- App signing via Play App Signing (Google manages final signing key)
- Target API level: must comply with Google Play's annual API level policy (currently API 34 minimum for new apps)
- Data safety form completed: data collected (name, email, location, payment info), data shared (payment processors), security practices
- Content rating: Everyone (IARC questionnaire)
- Staged rollout: 10% → 50% → 100% over 7 days with crash rate monitoring gate (<0.5% crash-free sessions required to proceed)
- In-app update API: mandatory update prompt when a breaking API version is deployed

Apple App Store:

- App Review Guidelines compliance: physical services exemption from IAP (3.1.1), no external payment links shown during checkout
- Privacy nutrition label: data used to track (none), data linked to identity (name, email, purchases), data not linked to identity (crash logs)
- Export compliance: uses standard encryption (HTTPS/TLS) — qualifies for ERN exemption
- Age rating: 4+
- Phased release: 7-day phased release enabled for all production submissions
- App Tracking Transparency (ATT): not required — ZikaBooking does not use cross-app tracking. ATT prompt not shown.
- Background modes: none declared (push handled by APNs, no background location)

Web / PWA deployment:

- Hosted on Vercel or AWS Amplify with global CDN edge nodes
- HTTPS enforced with HSTS preload. HTTPS Everywhere redirect.
- Content Security Policy (CSP) headers: script-src restricted to self + js.stripe.com + maps.googleapis.com
- Lighthouse PWA score target: 90+ on mobile (audited in CI pipeline)

- Web Vitals monitoring: Vercel Analytics or custom RUM solution. Alert on LCP regression > 300ms.
- Deployment: preview URL per pull request · staging environment · production blue/green with instant rollback
- Environment variables: NEXT_PUBLIC_* for client-safe config · server-only secrets for API keys (never in client bundle)

12. Provider dashboard

Every service provider — hotel operator, apartment host, or car fleet owner — has a dedicated dashboard scoped exclusively to their own listings and bookings. All financial figures displayed are net amounts after ZikaBooking's 5% service commission has been deducted. The provider always sees what they will actually receive, never the gross guest payment.

Important: all amounts shown in the provider dashboard are NET of ZikaBooking's 5% commission. A tooltip on every financial figure reads: "This is your earnings after ZikaBooking's 5% service fee. ZikaBooking retains the remainder."

12.1 Financial summary

Metric	Definition
Net revenue — all time	Sum of all completed payouts received, after 5% commission and taxes. Primary headline figure.
Net revenue — this month	Current calendar month earnings, net of commission.
Net revenue — last month	Prior month earnings for period-over-period comparison.
Pending payout	Earnings from bookings where check-in has occurred but T+24h payout not yet disbursed.
Total bookings — all time	Count of all confirmed bookings across all provider listings.
Bookings — this month	Confirmed bookings in the current calendar month.
Cancellation rate	Percentage of bookings cancelled in the last 90 days.
Average rating	Mean star rating across all listings, weighted by review count.

12.2 Upcoming bookings

A chronological list of all bookings with check-in date in the future (status = confirmed or active), across all provider listings.

Columns shown per booking row:

- Booking reference (ZIKA-XXXX-CC)
- Listing name
- Guest display name — first name + last initial only (e.g. "Amara D.") — never full surname, phone, or email
- Check-in date and check-out date
- Number of guests (hotels/apartments) or rental days (cars)
- Net payout amount (after 5% commission)

No guest contact details are shown. All guest-provider communication goes through the in-app messaging tool only (see section 13).

Sorting: By check-in date (default), listing name, or net payout amount. Filterable by listing and date range.

12.3 Available units

Per-listing availability overview showing for each listing: listing name, category (hotel/apartment/car), total units, units currently booked (confirmed bookings overlapping today), units held under active reservation locks, and units available now. For car fleets: per-vehicle breakdown. Clicking any listing opens its calendar.

12.4 Calendar — activate and deactivate offers by date

The provider calendar is the central availability management tool. It displays a monthly view per listing.

Date states — colour coded:

State	Colour	Meaning
Available	White (default)	Unit open to book
Booked	Green	Confirmed booking occupying this date
Held	Amber	Active 5-minute reservation lock
Blocked	Dark grey	Manually blocked by provider (maintenance, personal use)
External hold	Grey striped	Date blocked by imported iCal feed (Airbnb, Booking.com)
Promotion active	Red badge	ZikaBooking activity promotion running on this date
Provider offer	Orange badge	Provider-activated custom date-range discount

Calendar actions — provider selects a date range then chooses:

- **Block dates** — marks selected dates as unavailable. Immediately pushes a VEVENT BUSY block to the listing's iCal feed so Airbnb and Booking.com pick it up within 15 minutes.
- **Activate offer** — attaches a custom discount to selected dates. Provider sets: discount type (percentage or fixed), discount value, label text (max 6 chars, shown as an orange badge, e.g. "20%"). This is a provider-level offer distinct from admin activity promotions. Applies only to this listing on the selected dates. Discount auto-applied at checkout for guests booking those dates. Provider can deactivate at any time by selecting dates and clicking "Remove offer".
- **Unblock dates** — removes a manual block, making dates available again.

Sync status: A "Last synced" timestamp per connected OTA feed. Warning badge if sync has failed for more than 30 minutes.

Data scoping: Every dashboard query is filtered by provider_id at the API level. A provider can never see data belonging to another provider, even by manipulating query parameters.

13. In-app messaging tool

ZikaBooking includes a built-in messaging system for all communication between guests and providers. The platform enforces a strict no-contact-detail policy: phone numbers, email addresses, external URLs, social media handles, and any string matching a phone or email pattern are automatically detected and blocked from being sent. This protects both parties, keeps transactions on-platform, and prevents providers from diverting guests to off-platform bookings.

13.1 Conversation model

A conversation is created automatically when a booking is confirmed, or when a guest sends the first pre-booking enquiry message on a listing (if the listing has "Allow pre-booking enquiry" enabled). Each conversation is permanently linked to a booking_id or listing_id. Both parties are identified by their ZikaBooking display name and avatar — never anonymous.

Data model:

```
conversations: id, booking_id FK nullable, listing_id FK, guest_id FK, provider_id FK, created_at,
last_message_at, status (open/archived/flagged)

messages: id, conversation_id FK, sender_id FK, sender_role (guest/provider/support_agent), content
TEXT, content_filtered BOOLEAN, sent_at, read_at nullable, deleted_at nullable
```

13.2 Content filtering — no personal details

Every outbound message is filtered server-side before delivery. Blocked patterns include phone numbers, email addresses, external URLs, social media handles, and obfuscation attempts. The platform NEVER delivers contact details between guests and providers.

Blocked patterns:

- Phone numbers — international formats (+254..., 07XX XXX XXX, (555) 555-5555 etc.) via regex covering 50+ country formats
- Email addresses — RFC 5322 email pattern
- External URLs — http://, https://, www. prefix, or recognisable domain patterns (.com, .co.ke etc.)
- Social media handles — @username patterns, "WhatsApp", "Telegram", "Instagram", "Facebook" followed by username or number
- Deliberate obfuscation — numbers spelled out ("zero seven zero..."), spaces inserted in numbers, Unicode lookalike characters in domains

On detection: Message blocked and never delivered. Sender sees inline error: "Your message contains contact information. ZikaBooking requires all communication to stay in-app to protect both parties." Attempted text preserved in input field for editing. Blocked attempt logged to message_filter_log table.

Repeat violations: 3 blocked attempts within 24 hours → automated flag for trust & safety review. 5 violations → 48-hour messaging suspension with email notification.

13.3 Supported message types

- Text — plain, up to 2 000 characters per message
- Images — JPEG/PNG/WEBP, max 5 MB, uploaded via S3 pipeline (internal URL only, no contact details exposed)
- Quick replies — pre-defined chips for providers: "Check-in instructions", "Parking details", "WiFi password", "We'll be ready for you", "Unfortunately unavailable"

- Booking summary card — auto-sent by system on booking confirmation showing reference, dates, listing name, and net payout (provider) or total paid (guest)

13.4 Notifications

Push notification (FCM/APNs/Web Push) and email notification sent when a new message arrives and the recipient is not in the conversation.

Email: Contains sender display name, truncated message preview (first 80 chars), deep link to conversation. Never reveals sender contact details.

Frequency: Maximum one email per conversation per 30 minutes (batched). Push notifications are real-time.

Read receipts: Single tick = delivered to server. Double tick = read by recipient.

13.5 Support agent access

Support Agents can read any conversation for dispute resolution and trust & safety purposes. They can send messages labelled "ZikaBooking Support" with a shield badge. They cannot delete or modify existing messages. Messages are clearly distinguished from guest/provider messages.

13.6 Pre-booking enquiry

Listings with "Allow pre-booking enquiry" enabled allow guests (logged in + email-verified) to message a provider before booking. Pre-booking conversations are visible in the provider's messaging dashboard. If the guest proceeds to book, the conversation is automatically linked to the resulting booking.

13.7 Message retention & privacy

Messages retained for 24 months after associated booking checkout (or 24 months from conversation creation for pre-booking enquiries). After retention period, message content is deleted; metadata (sent_at, sender_role, content_filtered flag) kept for audit. Guests and providers can archive conversations but cannot permanently delete messages. GDPR data deletion requests fulfilled within 30 days.

14. Admin & sales agent — updated permissions

14.1 Admin can create country managers

Previously, only Super Admin could create Country Manager accounts. This permission is extended to the Admin role.

Updated Admin capabilities:

- Admin can create Country Manager accounts — must assign country scope at creation time
- Admin can create Sales Agent accounts
- Admin cannot create other Admin accounts (Super Admin only)
- Admin cannot create Super Admin accounts
- Admin cannot assign countries outside their own visibility scope

User creation form fields: Full name · email address · role (Country Manager or Sales Agent) · country scope (multi-select, required for Country Manager) · temporary password (auto-generated, emailed to new user with forced password-change on first login).

Audit trail: Every user creation logged to audit_log: creator user ID, creator role, new user ID, role assigned, country scope, timestamp.

Updated permission matrix row — "Create country managers":

Role	Old permission	New permission
Super Admin	Yes	Yes (unchanged)
Admin	No	Yes — can create Country Managers and Sales Agents
Country Manager	No	No (unchanged)
Sales Agent	No	No (unchanged)
Support Agent	No	No (unchanged)
Finance Agent	No	No (unchanged)

14.2 Sales agent — can answer booking requests

Sales Agents gain the ability to respond to request-to-book listings (where the host has enabled manual approval rather than instant confirm). This allows Sales Agents to act on behalf of a host when the host is unresponsive.

Scope: Sales Agents can only act on request-to-book requests within their assigned country scope. They cannot approve instant-confirm bookings, modify confirmed bookings, or issue refunds.

Available actions for Sales Agents on a pending request:

- Approve request — transitions booking from pending to confirmed, triggers payment charge, sends confirmation email
- Decline request — releases reservation lock, notifies guest with reason (Unavailable dates / Property under maintenance / Guest requirements not met / Other)
- Request more information — sends a message to guest via in-app messaging tool requesting clarification
- Escalate to host — sends notification to provider reminding them to respond within 24 hours; flags request in host dashboard

Visibility: Sales Agents see a "Pending requests" tab showing all request-to-book submissions in their country scope that have been waiting more than 2 hours without a host response. Prevents requests expiring without action.

New permission matrix row — "Answer booking requests":

Role	Permission
Super Admin	Yes — all countries
Admin	Yes — all countries
Country Manager	Yes — own countries
Sales Agent	Yes — own countries (request-to-book only)
Support Agent	No
Finance Agent	No

15. Commission management

ZikaBooking applies a platform service fee (commission) on every booking. Admins can override the default global rate at the country level, setting a different commission for specific countries or applying a new rate across all countries simultaneously. Commission adjustments take effect on new bookings only — confirmed bookings are never retroactively modified.

15.1 Commission model

Default rate: 5% platform-wide. Stored as `global_commission_rate` in the `platform_settings` table. This is the fallback rate for any country without a specific override.

Country override: A country-specific rate in the `country_commissions` table overrides the global default for all bookings where the listing's country code matches.

Effective rate resolution order:

- 1. Country-specific override (if exists for listing's country) → use that rate
- 2. Global default → use `global_commission_rate`

Binding moment: The commission rate active at the time the `PaymentIntent` is created server-side (payment step) is the binding rate for that booking. Lock creation time is not used. The applied rate is stored on the bookings record as `commission_rate_applied` and is immutable thereafter.

15.2 Access rights per role

Role	Global default	Country overrides	Bulk (all countries)	History & export
Super Admin	Read + write	Read + write (all)	Yes	Read + export
Admin	Read only	Read + write (own scope)	No	Read (own scope)
Country Manager	Read only	Read only (own countries)	No	Read (own countries)
Sales Agent	Read only	Read only (own countries)	No	No
Support Agent	Read only	Read only	No	No
Finance Agent	Read only	Read only (all)	No	Read + export (all)

15.3 Global commission adjustment (Super Admin only)

Fields:

- New global rate — decimal 0.00–50.00, max 2 decimal places, required
- Effective from — date, must be today or future (no backdating)
- Apply to all countries toggle — if ON, all existing country overrides are replaced by the new global rate (full reset); if OFF, country overrides remain unchanged and only the global default changes
- Reason — free text, max 500 chars, required (logged to audit trail)
- Notify providers toggle — if ON, email sent to all active providers with new rate, effective date, and reason

Confirmation step: Before saving, a modal displays: "You are about to change the global commission rate from X% to Y%, effective [date]. This will affect all new bookings [worldwide / in all countries]. Type CONFIRM to proceed." The confirmation word must be typed — button disabled until string matches exactly.

15.4 Country-specific commission adjustment

Available to Super Admin and Admin. The commission screen shows a table of all countries where ZikaBooking has active listings.

Country commission table columns: Country name · Country code · Number of active listings · Current commission rate · Rate source (Global default / Custom override) · Last modified date.

Adjust single country: Click country row → drawer opens with: country name (read-only), current rate (read-only), new rate field, effective date, reason (required), notify providers toggle.

Adjust multiple countries simultaneously: Select multiple countries via checkboxes → "Bulk edit selected" → set a single rate for all selected with one effective date and one reason.

Apply to all countries: Button visible to Super Admin only → opens global adjustment form with `apply_to_all_countries` pre-toggled ON.

Remove country override: Deletes the country-specific override, reverting that country to the global default. Confirmation dialog: "Removing this override will revert [Country] to the global default rate of X%."

15.5 Validation rules

- Rate must be decimal between 0.00 and 50.00 — values above 50% rejected at API level
- Rate of 0.00 is permitted (free tier, promotional markets, test environments)
- Maximum 2 decimal places — 12.50 valid, 12.555 rejected
- Effective date cannot be in the past
- Rate change identical to current rate rejected with "This country already has this commission rate"
- A country cannot have more than one pending future rate change at a time — new one supersedes previous pending one, with warning shown to admin

15.6 Scheduling & effective date promotion

Commission changes can be scheduled for a future date. The platform stores both `current_rate` and any `pending_rate` with its `effective_from` date. A background job runs nightly at 00:01 UTC and promotes pending rates whose `effective_from` date has been reached to `current_rate`. The old `current_rate` is moved to `commission_history`.

15.7 Commission history & audit

A complete, immutable history of every commission change is maintained in the `commission_history` table. Viewable under Settings → Commission → History. Filterable by country, date range, and changed-by user.

commission_history columns: `id` · `scope` (global/country) · `country_code` (null if global) · `old_rate` · `new_rate` · `effective_from` · `changed_by` (admin user ID) · `changed_by_role` · `reason` · `apply_to_all` (boolean) · `providers_notified` (boolean) · `created_at`.

Export: CSV export available to Super Admin and Finance Agent.

15.8 Provider notification email

When `notify_providers` is ON, email sent to all active providers in affected countries via SendGrid within 1 hour (batched to respect rate limits).

Subject: "Important: ZikaBooking commission rate update for [Country / All markets]"

Body: Current rate → New rate · Effective date · Admin reason · Link to provider dashboard earnings · Note: "This change applies to new bookings confirmed on or after [effective date]. Your existing confirmed bookings are not affected."

15.9 Impact on billing formula

The billing calculation in section 5.4 is updated to use a dynamic commission rate:

```
commission_rate = get_effective_rate(listing.country_code, payment_intent_created_at)
base_amount = nights_or_days × unit_price
discount = best(promotion_discount, voucher_discount)
subtotal = base_amount - discount
service_fee = CEILING(subtotal × commission_rate / 100, 2 dp)
taxes = subtotal × country_tax_rate
total = subtotal + service_fee + taxes
```

commission_rate_applied is stored on every bookings record at creation time as DECIMAL(5,2), immutable after creation, so historical bookings always reflect the rate in effect when they were made.

16. Non-functional requirements

Performance	Search results < 1.5s. Booking confirmation < 3s. Reservation lock creation < 500ms.
Scalability	Horizontal scaling. CDN for images and static assets. Database read replicas for search queries.
Security	PCI-DSS compliance via Stripe vault (no raw card data stored). JWT auth with refresh token rotation. Ra
Availability	99.9% uptime SLA. Auto-failover for database and payment services. Graceful degradation when third-p
Localisation	UI in English, French, Arabic, Portuguese, Swahili at launch. RTL support for Arabic. Date format and tir
Accessibility	WCAG 2.1 AA compliance. Screen-reader support. Keyboard navigation throughout booking flow.
Payments	No payment credentials on ZikaBooking servers. All Stripe keys in environment variables. Tara API keys
Audit	Every admin action logged to audit_log table. Payment events to payment_events table. Email sends to

17. Technology stack

Frontend	React Native (iOS + Android) · Next.js (web)
Backend	Node.js microservices · REST + GraphQL
Database	PostgreSQL (bookings, users) · Redis (sessions, search cache, reservation locks) · Elasticsearch (listing
Infrastructure	AWS or GCP · Docker / Kubernetes · S3/GCS for PDF and photo storage
Notifications	Firebase Cloud Messaging (push) · SendGrid (transactional email)
Maps	Google Maps Platform (Geocoding API, Places API, Maps JS)
Payments	Stripe (Connect Express, Webhooks) · Tara API (mobile money aggregator)
Calendar sync	RFC 5545 iCal poller · Phase 2: Airbnb Connect API, Booking.com Connectivity API
PDF generation	pdf-lib or Puppeteer (server-side) · stored in S3 · attached to SendGrid emails
Security	bcrypt (passwords) · crypto.randomBytes (tokens) · SHA-256 (token hashing) · HMAC (webhook validation
FX rates	Third-party FX rates API · updated hourly · informational display only

18. Data model summary

users	id, first_name, last_name, email, password_hash, status (pending_verification/active/suspended/banned), user_type (guest/provider), email_verified, email_verified_at, current_tier, loyalty_points, created_at
verification_tokens	id, user_id FK, token_hash CHAR(64), token_type, expires_at, used, used_at, invalidated_reason, created_at
listings	id, provider_id FK, category (hotel/apartment/car), name, status (draft/pending_review/active/auto_suspended/permanently_banned), address, lat, lng, town, country, unit_price, currency, consecutive_negative_count, auto_suspension_count, voucher_pdf_url
reservation_locks	id, listing_id FK, guest_id FK, session_token, dates_from, dates_to, unit_count_held, payment_rail, status (active/completed/expired/released), created_at, expires_at, completed_at
bookings	id, listing_id FK, guest_id FK, reference ZIKA-XXXX-CC, booking_seq, country_code, nights_or_days, unit_price, base_amount, discount_type, discount_amount, subtotal, service_fee, commission_rate_applied DECIMAL(5,2), tax_rate, tax_amount, total_amount, currency, payment_method, payment_tx_id, payment_date, voucher_pdf_url, status
payments	id, booking_id FK, rail (tara/stripe), external_id, amount, currency, status (pending/completed/failed/refunded), created_at
payouts	id, booking_id FK, host_id FK, rail, external_id, amount, currency, status, scheduled_at, completed_at
vouchers	id, code VARCHAR(20) UNIQUE, title, activity_scope TEXT[], discount_type, discount_value, valid_from, valid_until, usage_limit_total, usage_limit_per_guest, current_usage_count, status, applicable_tiers TEXT[], tier_mode, auto_assign, country_scope, created_by FK
activity_promotions	id, activity ENUM(hotel/apartment/car), label_text VARCHAR(6), discount_type, discount_value, valid_from, valid_until, apply_to_booking, banner_title, status (scheduled/active/paused/expired/superseded), country_scope
reviews	id, booking_id FK, listing_id FK, guest_id FK, rating SMALLINT 1-5, comment TEXT, created_at, fraud_flagged
listing_suspensions	id, listing_id FK, trigger_type, triggering_review_ids UUID[], suspended_at, status (open/resolved/escalated/permanent), assigned_to FK, resolution, agent_note, escalation_level, sla_deadline
admin_users	id, name, email, role (super_admin/admin/country_manager/sales/support/finance), country_scope CHAR(2)[], totp_enabled, created_at
audit_log	id, user_id FK, role, action, target_type, target_id, old_value, new_value, ip_address, timestamp – immutable, 7-year retention
email_log	id, user_id FK, booking_id FK, type, recipient, status (sent/failed/bounced), sendgrid_message_id, sent_at
channel_feeds	id, listing_id FK, platform_name, ical_import_url, ical_export_token, last_synced_at, status, blocked_dates_count
conversations	id, booking_id FK nullable, listing_id FK, guest_id FK, provider_id FK, created_at, last_message_at, status (open/archived/flagged)

messages	id, conversation_id FK, sender_id FK, sender_role (guest/provider/support_agent), content TEXT, content_filtered BOOLEAN, sent_at, read_at nullable, deleted_at nullable
message_filter_log	id, conversation_id FK, sender_id FK, blocked_content_type (phone/email/url/social/obfuscation), attempted_at, violation_count
provider_offers	id, listing_id FK, provider_id FK, dates_from DATE, dates_to DATE, discount_type (percentage/fixed), discount_value DECIMAL, label_text VARCHAR(6), status (active/inactive), created_at
country_commissions	id, country_code CHAR(2) UNIQUE, current_rate DECIMAL(5,2), pending_rate DECIMAL(5,2) nullable, pending_effective_from DATE nullable, last_modified_at, last_modified_by FK
commission_history	id, scope ENUM(global/country), country_code CHAR(2) nullable, old_rate DECIMAL(5,2), new_rate DECIMAL(5,2), effective_from DATE, changed_by FK, changed_by_role, reason TEXT, apply_to_all BOOLEAN, providers_notified BOOLEAN, created_at – immutable

19. Process map — domain overview

ZikaBooking is structured around four interconnected domains. Each domain encapsulates a distinct set of processes and is linked to the others via shared data entities and event flows.

Domain 1 — User & identity	Registration · email verification (SHA-256 token, 24h TTL, single-use) · OAuth fast path (Google/Apple, pre-verified) · Account status lifecycle (pending → active → suspended/banned) · JWT sessions (8h inactivity timeout) · 2FA for admin panel · Profile (name, photo, payment methods, loyalty points, tier)
Domain 2 — Listings & inventory	Hotel creation (manual approval + star rating by admin) · Apartment creation (auto-activated) · Car rental creation (auto-activated) · Shared geocoding engine → live map pin · Amenity grid (5 categories, custom add) · Channel manager: iCal two-way sync with Airbnb + Booking.com, poll every 15 min, UID dedup · Auto-suspension: 2× consecutive 1★–2★ reviews on apts/cars → listing suspended, agent review required
Domain 3 — Booking & payment	5-minute reservation lock (atomic SETNX, colour-coded timer, 2-min grace window) · Two payment rails: Tara (mobile money, Africa) and Stripe (global, 135+ currencies) · Billing: nights×price – best discount + 5% service fee (CEILING) + country VAT → total · ZIKA-XXXX-CC reference (DB sequence, immutable) · Confirmation email (guest + host) + PDF voucher (A4, S3-stored, attached) · Cancellation with policy-aware refund
Domain 4 — Promotions, vouchers, admin & post-stay	Personal vouchers (activity-scoped, tier-locked, wallet, dismissable Best Offer banner) · Activity promotions (permanent red banner + badge on all listing cards, one per category, best discount wins) · Loyalty programme (Bronze/Silver/Gold/Diamond, auto-assign on upgrade) · Admin panel (6 roles, country manager scoping, hotel accreditation queue, unblock queue, audit log) · Post-stay: review → AfriPoints → payout T+24h → cancellation handling

External system integrations:

Google Maps Platform	Geocoding API (address → lat/lng), Places API (autocomplete), Maps JS (embedded map pin)
Airbnb + Booking.com	iCal export/import feeds · Phase 2: direct partner APIs
SendGrid	Transactional email delivery · webhook events for bounce/spam tracking
Tara API	Unified mobile money aggregator · STK push · disbursements · HMAC webhooks
Stripe	PaymentIntent API · Elements (PCI-DSS) · Connect Express · Webhooks
FX rates API	Live exchange rates (hourly) · informational display only · charges in listing base currency
Firebase Cloud Messaging	Push notifications (booking confirmations, timer alerts, voucher auto-assign, payout sent)
S3 / GCS	PDF voucher storage · listing photo storage · CDN delivery